

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 22
Design of FIR Filters Using the Windowing Method

III B.Tech EEE — Core Exam Preparation Guide

1 Core Concept & Principle of Windowing

1.1 The Truncation Problem & Ideal Impulse Response Derivation

An ideal brick-wall lowpass filter with cutoff frequency ω_c has the desired frequency response:

$$H_d(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases} \quad (1)$$

The continuous impulse response $h_d[n]$ is obtained via the ****Inverse DTFT****:

$$\begin{aligned} h_{d,\text{LP}}[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} 1 \cdot e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{-\omega_c}^{\omega_c} = \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{2\pi jn} \end{aligned} \quad (2)$$

Applying Euler's sine formula $e^{j\theta} - e^{-j\theta} = 2j \sin \theta$:

$$h_{d,\text{LP}}[n] = \frac{2j \sin(\omega_c n)}{2\pi jn} = \frac{\sin(\omega_c n)}{\pi n}, \quad -\infty < n < \infty \quad (3)$$

At $n = 0$, applying L'Hôpital's rule: $h_{d,\text{LP}}[0] = \lim_{n \rightarrow 0} \frac{\sin(\omega_c n)}{\pi n} = \frac{\omega_c}{\pi}$. This impulse response is doubly unphysical: it has ****infinite duration (IIR)**** and is ****non-causal**** (extends to $-\infty$).

To realize a physical causal FIR filter of length N , we:

1. Shift the center of symmetry to $\tau = \alpha = \frac{N-1}{2}$ to ensure causality and linear phase.
2. Multiply $h_d[n]$ pointwise by a finite-duration window sequence $w[n]$ ($0 \leq n \leq N-1$):

$$\mathbf{h}[\mathbf{n}] = \mathbf{h}_d[\mathbf{n}] \cdot \mathbf{w}[\mathbf{n}], \quad 0 \leq n \leq N-1 \quad (4)$$

1.2 Frequency-Domain Convolution & Gibbs Phenomenon

By the DTFT modulation property, time-domain multiplication corresponds to periodic convolution in frequency:

$$H(e^{j\omega}) = \frac{1}{2\pi} H_d(e^{j\omega}) \circledast W(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta \quad (5)$$

- **Mainlobe Width of $W(e^{j\omega})$:** Spreads out the ideal vertical transition into a gradual transition band $\Delta\omega$.
- **Sidelobes of $W(e^{j\omega})$:** Cause oscillations (ringing) in the passband and stopband.
- **Gibbs Phenomenon:** Abrupt rectangular truncation results in a fixed 8.95% overshoot (≈ -21 dB peak sidelobe) near the discontinuity, which **cannot be reduced by increasing N** . Smoother, tapered windows are necessary to suppress sidelobes.

2 Master Window Summary Table (Formulas & Specs)

Table 1: Characteristics of Standard FIR Filter Window Functions

Window Name	Time-Domain Equation $w[n]$ ($0 \leq n \leq N-1$)	Transition Width $\Delta\omega$	Peak Sidelobe	Min. Attenuation A_s
Rectangular	$w[n] = 1$	$\frac{4\pi}{N}$ (or $\frac{1.8\pi}{N}$)	-13 dB	21 dB
Bartlett (Triangular)	$w[n] = 1 - \frac{2 n-(N-1)/2 }{N-1}$	$\frac{8\pi}{N}$ (or $\frac{6.4\pi}{N}$)	-27 dB	25 dB
Hann (Hanning)	$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right)$	$\frac{8\pi}{N}$ (or $\frac{6.2\pi}{N}$)	-32 dB	44 dB
Hamming	$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$	$\frac{8\pi}{N}$ (or $\frac{6.6\pi}{N}$)	-43 dB	53 dB
Blackman	$w[n] = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right)$	$\frac{12\pi}{N}$ (or $\frac{11\pi}{N}$)	-58 dB	74 dB

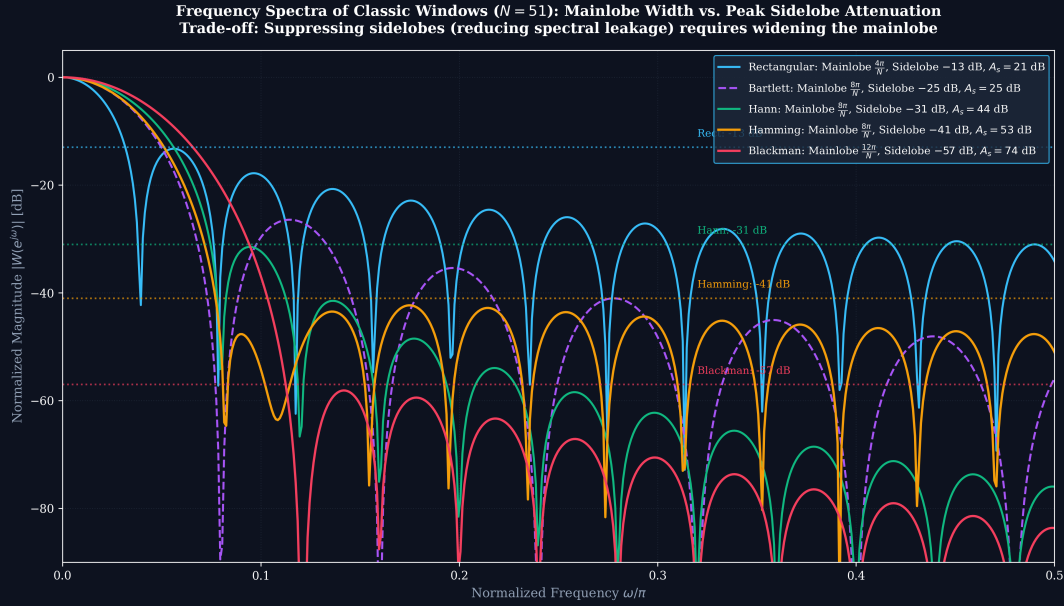


Figure 1: Normalized frequency magnitude spectra of classic window functions ($N = 51$). Note the fundamental trade-off: narrower mainlobe (faster transition roll-off) vs. lower sidelobes (higher stopband rejection).

Key Engineering Trade-offs & Exam Rules:

- Mainlobe Width vs. Sidelobe Attenuation Trade-off:** As the window tapers more smoothly to zero at both ends (from Rectangular $\rightarrow$ Bartlett $\rightarrow$ Hann $\rightarrow$ Hamming $\rightarrow$ Blackman), the stopband attenuation improves dramatically from 21 dB to 74 dB, but the mainlobe width expands from $4\pi/N$ to $12\pi/N$. A wider mainlobe requires a larger filter length N to meet the same transition bandwidth $\Delta\omega$.
- Why Hamming is the Universal Exam Default:** Hamming achieves **53 dB** stopband attenuation with the same mainlobe width ($\approx 8\pi/N$) as the Hann window (44 dB), because its nonzero pedestal (0.08) cancels the first sidelobe.
- Why N Cannot Reduce Sidelobes:** In fixed windows, increasing N compresses the frequency axis (narrowing the mainlobe), but the **relative height of the sidelobes in dB remains constant**. Only choosing a different window type improves stopband attenuation A_s .

3 Ideal Shifted Impulse Response Formulations

To design a causal, linear-phase FIR filter of length N , the ideal frequency response incorporates a linear phase delay $\tau = \alpha = \frac{N-1}{2}$:

$$H_d(e^{j\omega}) = \begin{cases} 1 \cdot e^{-j\omega\tau}, & |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases} \quad (6)$$

Evaluating the Inverse DTFT with all intermediate integral steps:

$$\begin{aligned} h_d[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\tau} e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\tau)} d\omega = \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\tau)}}{j(n-\tau)} \right]_{-\omega_c}^{\omega_c} \\ &= \frac{e^{j\omega_c(n-\tau)} - e^{-j\omega_c(n-\tau)}}{2\pi j(n-\tau)} = \frac{2j \sin(\omega_c(n-\tau))}{2\pi j(n-\tau)} = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}, \quad n \neq \tau \end{aligned} \quad (7)$$

At the center of symmetry $n = \tau$, applying L'Hôpital's rule:

$$h_d[\tau] = \lim_{n \rightarrow \tau} \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)} = \lim_{n \rightarrow \tau} \frac{\omega_c \cos(\omega_c(n-\tau))}{\pi \cdot 1} = \frac{\omega_c}{\pi} \quad (8)$$

Highpass, bandpass, and bandstop filters follow directly from linear combinations of ideal lowpass filters:

- **Highpass:** By spectral inversion, $H_{d,HP}(e^{j\omega}) = e^{-j\omega\tau} - H_{d,LP}(e^{j\omega}) \implies h_{d,HP}[n] = \delta[n - \tau] - h_{d,LP}[n]$.
- **Bandpass:** As difference of two lowpass filters, $h_{d,BP}[n] = h_{d,LP2}[n] - h_{d,LP1}[n]$.
- **Bandstop:** $h_{d,BS}[n] = \delta[n - \tau] - h_{d,BP}[n]$.

Table 2: Closed-Form Formulas for Ideal Shifted Filters $h_d[n]$

Filter Type	For $n \neq \tau$	Center Tap $n = \tau$
Lowpass (LPF)	$h_d[n] = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = \frac{\omega_c}{\pi}$
Highpass (HPF)	$h_d[n] = -\frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = 1 - \frac{\omega_c}{\pi}$
Bandpass (BPF)	$h_d[n] = \frac{\sin(\omega_{c2}(n-\tau)) - \sin(\omega_{c1}(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = \frac{\omega_{c2} - \omega_{c1}}{\pi}$
Bandstop (BSF)	$h_d[n] = \frac{\sin(\omega_{c1}(n-\tau)) - \sin(\omega_{c2}(n-\tau))}{\pi(n-\tau)}$	$h_d[\tau] = 1 - \frac{\omega_{c2} - \omega_{c1}}{\pi}$

3.1 Linear-Phase FIR Filter Constraints & Symmetry Types

To preserve exact linear phase without phase distortion, FIR impulse responses must satisfy symmetry:

- **Type 1 (Symmetric, N Odd):** Impulse response satisfies $h[n] = h[N-1-n]$ with integer delay $\tau = \frac{N-1}{2}$. The center sample is at an integer index $n = \tau$. Type 1 filters can realize **all four filter classes** (Lowpass, Highpass, Bandpass, and Bandstop).
- **Type 2 (Symmetric, N Even):** Symmetrical about a half-integer $\tau = \frac{N-1}{2}$. This symmetry strictly forces a null at $\omega = \pi$: $H(e^{j\pi}) = 0$. Therefore, **an even-length symmetric filter can NEVER be used for Highpass or Bandstop filters** where transmission at $\omega = \pi$ is required!
- **Design Takeaway for Exams:** Always verify that N is chosen **odd** for highpass or bandstop designs, guaranteeing full passband transmission at Nyquist ($\omega = \pi$).

4 FIR Filter Specifications & 5-Step Window Design Procedure

4.1 Filter Tolerance Specifications & Frequency Bands

To design a physical FIR filter using the windowing method, the ideal brick-wall response must be approximated within acceptable engineering tolerance bands, as illustrated in Figure 2:

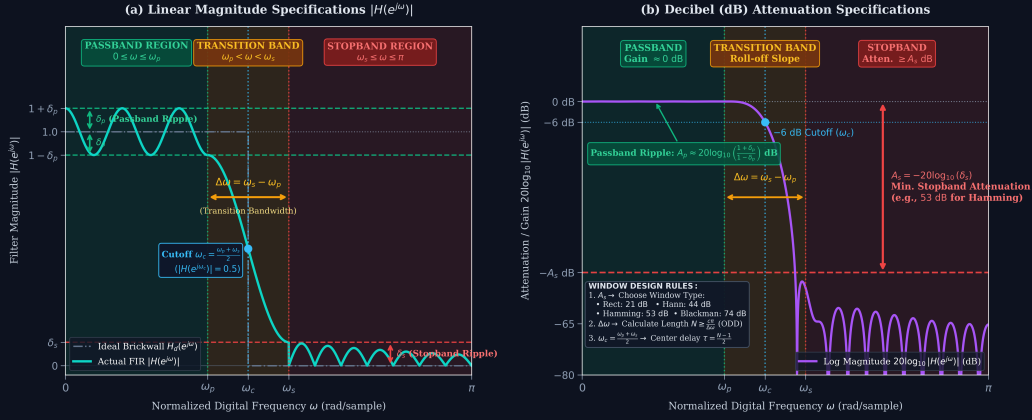


Figure 2: FIR Filter Tolerance Specifications and Parameter Mapping: (a) Linear magnitude $|H(e^{j\omega})|$ displaying passband ripple δ_p , stopband ripple δ_s , transition bandwidth $\Delta\omega = \omega_s - \omega_p$, and -6 dB cutoff frequency $\omega_c = (\omega_p + \omega_s)/2$; (b) Decibel attenuation showing minimum stopband attenuation $A_s = -20 \log_{10}(\delta_s)$ dB and systematic window design mapping.

- **Passband** ($0 \leq \omega \leq \omega_p$): Band where desired frequencies pass. The magnitude stays bounded within $1 - \delta_p \leq |H(e^{j\omega})| \leq 1 + \delta_p$. Peak-to-peak ripple is $2\delta_p$, and passband ripple in decibels is $A_p \approx 20 \log_{10} \frac{1+\delta_p}{1-\delta_p}$ dB.
- **Transition Band** ($\omega_p < \omega < \omega_s$): Finite roll-off band connecting passband and stopband. The **transition bandwidth** $\Delta\omega = \omega_s - \omega_p$ is governed by the window mainlobe width ($\Delta\omega \approx \frac{c\pi}{N}$), which **dictates the minimum filter length** N .
- **Cutoff Frequency** (ω_c): The ideal transition midpoint where gain drops to 0.5 (-6 dB attenuation): $\omega_c = \frac{\omega_p + \omega_s}{2}$.
- **Stopband** ($\omega_s \leq \omega \leq \pi$): Band where unwanted frequencies are suppressed: $|H(e^{j\omega})| \leq \delta_s$. The **minimum stopband attenuation** $A_s = -20 \log_{10}(\delta_s)$ dB is governed by the window peak sidelobe, which **dictates the choice of window type** (independent of N).

4.2 The 5-Step Systematic Design Procedure

1. **Step 1 — Identify Stopband Attenuation A_s :** From δ_s , calculate $A_s = -20 \log_{10}(\delta_s)$ dB. Select the window from Table 1 whose peak sidelobe attenuation meets or exceeds A_s .
2. **Step 2 — Calculate Filter Length N :** From $\Delta\omega = \omega_s - \omega_p$, determine $N \geq \frac{c\pi}{\Delta\omega}$ from Table 1. Round up to the next odd integer to ensure Type 1 linear phase (integer delay $\tau = (N - 1)/2$).
3. **Step 3 — Determine Cutoff Frequency ω_c :** Set $\omega_c = \frac{\omega_p + \omega_s}{2}$.
4. **Step 4 — Formulate Shifted Ideal Impulse Response $h_d[n]$:** Evaluate $h_d[n]$ for $n = 0, 1, \dots, N - 1$ using Table 2 with delay $\tau = \frac{N-1}{2}$.
5. **Step 5 — Apply Window Multiplication:** Compute final filter coefficients: $h[n] = h_d[n] \cdot w[n]$.

5 Standard Solved Exam Example 1: Lowpass Filter Design

Problem: Design a linear-phase FIR lowpass filter with cutoff frequency $\omega_c = 0.5\pi$ rad/sample and length $N = 7$ using a **Hamming window**. Determine impulse response $h[n]$, system function $H(z)$, and verify frequency response at $\omega = 0$ (DC) and $\omega = \pi$ (Nyquist).

Solution:

Step 1: Symmetry Delay Parameter

For length $N = 7$ (odd), the center of symmetry is $\tau = \alpha = \frac{N-1}{2} = \frac{7-1}{2} = \mathbf{3}$ (integer delay $\implies$ Type 1 FIR).

Step 2: Ideal Shifted Impulse Response Evaluation $h_d[n]$

From Table 2, the shifted ideal LPF formula with $\omega_c = 0.5\pi$ is $h_d[n] = \frac{\sin(0.5\pi(n-3))}{\pi(n-3)}$ for $n \neq 3$, and center tap $h_d[3] = \frac{\omega_c}{\pi} = \frac{0.5\pi}{\pi} = \mathbf{0.5000}$. Evaluating taps:

$$\begin{aligned} n = 2 : h_d[2] &= \frac{\sin(0.5\pi(-1))}{-\pi} = \frac{-1}{-\pi} = \frac{1}{\pi} \approx \mathbf{0.3183} & n = 1 : h_d[1] &= \frac{\sin(-\pi)}{-2\pi} = \mathbf{0.0000} \\ n = 0 : h_d[0] &= \frac{\sin(-1.5\pi)}{-3\pi} = \frac{1}{-3\pi} = -\frac{1}{3\pi} \approx \mathbf{-0.1061} \end{aligned} \quad (9)$$

By even symmetry about $\tau = 3$ ($h_d[3+k] = h_d[3-k]$): $h_d[4] = h_d[2] = 0.3183$, $h_d[5] = h_d[1] = 0.0000$, and $h_d[6] = h_d[0] = -0.1061$.

Step 3: Hamming Window Weights $w[n]$

The formula for a length-7 Hamming window ($0 \leq n \leq 6$) is $w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{6}\right) = 0.54 - 0.46 \cos\left(\frac{\pi n}{3}\right)$:

$$\begin{aligned} n = 3 : w[3] &= 0.54 - 0.46(-1) = \mathbf{1.0000} & n = 2 : w[2] &= 0.54 - 0.46(-0.5) = \mathbf{0.7700} \\ n = 1 : w[1] &= 0.54 - 0.46(0.5) = \mathbf{0.3100} & n = 0 : w[0] &= 0.54 - 0.46(1) = \mathbf{0.0800} \end{aligned} \quad (10)$$

By window symmetry: $w[4] = w[2] = 0.7700$, $w[5] = w[1] = 0.3100$, $w[6] = w[0] = 0.0800$.

Step 4: Windowing Multiplication $h[n] = h_d[n] \cdot w[n]$

Multiplying corresponding values tap-by-tap:

$$\begin{aligned} h[3] &= 0.5000 \times 1.0000 = \mathbf{0.5000}, & h[2] &= h[4] = 0.31831 \times 0.7700 = \mathbf{0.2451}, \\ h[1] &= h[5] = 0.00000 \times 0.3100 = \mathbf{0.0000}, & h[0] &= h[6] = -0.10610 \times 0.0800 = \mathbf{-0.0085}. \end{aligned}$$

$$h[n] = \{-\mathbf{0.0085}, \mathbf{0.0000}, \mathbf{0.2451}, \mathbf{0.5000}, \mathbf{0.2451}, \mathbf{0.0000}, -\mathbf{0.0085}\} \quad (11)$$

Step 5: System Function $H(z)$ and Frequency Response Verification

Transfer function: $H(z) = -0.0085 + 0.2451z^{-2} + 0.5000z^{-3} + 0.2451z^{-4} - 0.0085z^{-6}$.

Factoring linear phase delay $e^{-j3\omega}$ on the unit circle $z = e^{j\omega}$:

$$\begin{aligned} H(e^{j\omega}) &= e^{-j3\omega} [h[3] + 2h[2] \cos(\omega) + 2h[1] \cos(2\omega) + 2h[0] \cos(3\omega)] \\ &= e^{-j3\omega} [0.5000 + 2(0.2451) \cos(\omega) + 0 - 2(0.0085) \cos(3\omega)] \\ &= e^{-j3\omega} [0.5000 + 0.4902 \cos(\omega) - 0.0170 \cos(3\omega)] \end{aligned} \quad (12)$$

Critical Frequency Verification:

- **DC Gain** ($\omega = 0$): With $\cos(0) = 1$: $H(e^{j0}) = 0.5000 + 0.4902(1) - 0.0170(1) = \mathbf{0.9732} \approx 1.0$ (ideal passband transmission).
- **Nyquist Gain** ($\omega = \pi$): With $\cos(\pi) = \cos(3\pi) = -1$: $H(e^{j\pi}) = e^{-j3\pi} [0.5000 - 0.4902 + 0.0170] = (-1) \cdot [0.0268] = \mathbf{-0.0268}$.
Stopband attenuation at π : $|H(e^{j\pi})| = 0.0268 \implies 20 \log_{10}(0.0268) \approx \mathbf{-31.44}$ dB, verifying strong lowpass attenuation with only 7 taps.

6 Solved Exam Example 2: Highpass Filter Design

Problem: Design a 5-tap linear-phase FIR highpass filter with cutoff frequency $\omega_c = 0.5\pi$ rad/sample using a **Rectangular window**. Determine $h[n]$ and verify its frequency response at $\omega = 0$ and $\omega = \pi$.

Solution:

Step 1: Symmetry Delay Parameter

Filter length $N = 5 \implies \tau = \frac{5-1}{2} = 2$.

Step 2: Ideal Shifted Highpass Impulse Response $h_d[n]$

From Table 2, the shifted ideal highpass formula is:

$$h_d[n] = \begin{cases} 1 - \frac{\omega_c}{\pi} = 1 - \frac{0.5\pi}{\pi} = 0.5000, & n = \tau = 2 \\ -\frac{\sin(0.5\pi(n-2))}{\pi(n-2)}, & n \neq 2 \end{cases} \quad (13)$$

Evaluating each tap explicitly:

$$\begin{aligned} n = 2 : h_d[2] &= 1 - 0.5 = \mathbf{0.5000} \\ n = 1 : h_d[1] &= -\frac{\sin(0.5\pi(-1))}{-\pi} = -\frac{-1}{-\pi} = -\frac{1}{\pi} \approx \mathbf{-0.3183} \\ n = 0 : h_d[0] &= -\frac{\sin(0.5\pi(-2))}{-2\pi} = -\frac{0}{-2\pi} = \mathbf{0.0000} \end{aligned} \quad (14)$$

By even symmetry about $\tau = 2$: $h_d[3] = h_d[1] = -0.3183$ and $h_d[4] = h_d[0] = 0.0000$.

Step 3: Rectangular Window & Multiplication

For a rectangular window, $w[n] = 1$ for $0 \leq n \leq 4$. Therefore $h[n] = h_d[n]$:

$$\mathbf{h[n] = \{0.0000, -0.3183, 0.5000, -0.3183, 0.0000\}} \quad (15)$$

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Step 4: Frequency Response Verification

$$H(e^{j\omega}) = e^{-j2\omega}[h[2] + 2h[1]\cos(\omega) + 2h[0]\cos(2\omega)] = e^{-j2\omega}[0.5000 - 0.6366\cos(\omega)] \quad (16)$$

- **Stopband at DC ($\omega = 0$):** $H(e^{j0}) = 0.5000 - 0.6366(1) = \mathbf{-0.1366} \approx 0$ (attenuated).
- **Passband at Nyquist ($\omega = \pi$):** $H(e^{j\pi}) = e^{-j2\pi}[0.5000 - 0.6366(-1)] = 1 \cdot [0.5000 + 0.6366] = \mathbf{1.1366} \approx 1$ (passed).

7 Essential Exam Checklist & Traps

- **Center Tap Formula:** For LPF, $h_d[\tau] = \omega_c/\pi$. Do NOT attempt $\frac{\sin(0)}{0}$ without taking the limit!
- **Highpass Inversion:** For HPF, $h_d[\tau] = 1 - \omega_c/\pi$. All other taps are the negative of the LPF taps.
- **Window Selection Rule:** If the specification requires $A_s = 50$ dB, a Hann window ($A_s = 44$ dB) is insufficient. You **must** select **Hamming** ($A_s = 53$ dB).
- **Argument in Window Cosine:** Remember the denominator is $N - 1$, not N : $\cos\left(\frac{2\pi n}{N-1}\right)$.